

Google Play Store Analytics Dashboard

1. Project Title

Google Play Store Analytics Dashboard

2. Introduction

This project focuses on analyzing Google Play Store application data to uncover trends, patterns, and insights related to app categories, installs, ratings, pricing, and user sentiments. The dashboard was developed using Power BI to provide interactive visualizations and meaningful business insights for developers, analysts, and businesses.

3. Project Objectives

- Analyze Google Play Store app market trends.
- Explore app categories and user preferences.
- Compare free and paid applications.
- Study relationships between ratings, installs, and reviews.
- Perform sentiment analysis on user reviews.
- Build an interactive Power BI dashboard for data-driven insights.

4. Dataset Description

Dataset Source: Kaggle – Android App Market on Google Play Store.

The dataset contains information about Android applications such as:

- App Name
- Category
- Rating
- Reviews
- Installs
- Price
- Type (Free/Paid)
- Content Rating
- Genres
- User Sentiment Reviews

5. Tools & Technologies Used

- Power BI – Dashboard creation and visualization
- Python – Data cleaning and preprocessing
- Pandas & NumPy – Data manipulation

- Matplotlib & Seaborn – Visualization support
- DAX – KPI calculations and measures
- Jupyter Notebook – Data analysis environment

6. Data Cleaning & Preparation

- Removed duplicate records.
- Handled missing values.
- Converted installs and price columns into numeric format.
- Removed symbols such as '+', ',', and '\$'.
- Standardized category names and app data.
- Processed sentiment review data.
- Created calculated columns and measures for dashboard analysis.

7. Key KPIs (Using DAX)

Total Apps = COUNT(Apps[App])

Average Rating = AVERAGE(Apps[Rating])

Total Installs = SUM(Apps[Installs])

Total Reviews = SUM(Apps[Reviews])

8. Dashboard Overview

The dashboard includes:

- KPI Cards for total apps, installs, and reviews.
- Ratings vs Installs scatter plot.
- Sentiment analysis visualizations.
- App ratings distribution.
- Free vs Paid app comparison.
- Apps by category analysis.
- Top installed apps analysis.
- Interactive slicers for category and sentiment filtering.

9. Key Insights & Findings

- Free apps dominate the Play Store market.
- Family and Game categories contain the highest number of apps.
- Highly rated apps generally achieve more installs.
- Positive sentiments dominate user reviews.
- Most apps have ratings between 4.0 and 4.5.
- Popular apps generate significantly higher review counts.

10. Business Impact

- Helps developers understand successful app characteristics.
- Supports business decisions using market insights.
- Identifies high-performing categories and user trends.
- Improves customer understanding through sentiment analysis.
- Assists in app pricing and marketing strategies.

11. Conclusion

The project successfully demonstrates how data analytics and visualization techniques can be used to analyze the Google Play Store ecosystem. The dashboard provides meaningful insights into app popularity, ratings, pricing strategies, and customer sentiments, enabling data-driven decision-making.

12. Future Enhancements

- Add machine learning models for app success prediction.
- Integrate real-time Play Store data.
- Perform advanced NLP-based sentiment analysis.
- Build predictive analytics dashboards.
- Deploy dashboard to cloud platforms for live access.

13. Dashboard Screenshots

Main Google Play Store Analytics Dashboard



Sentiment Analysis Dashboard

